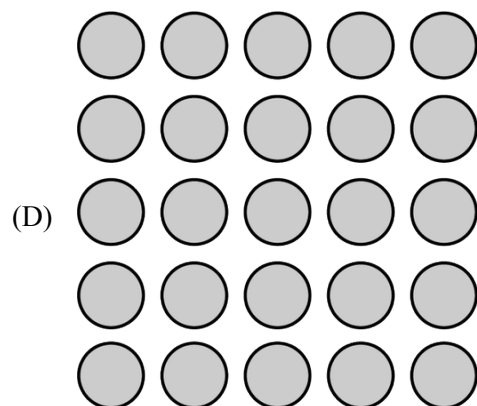
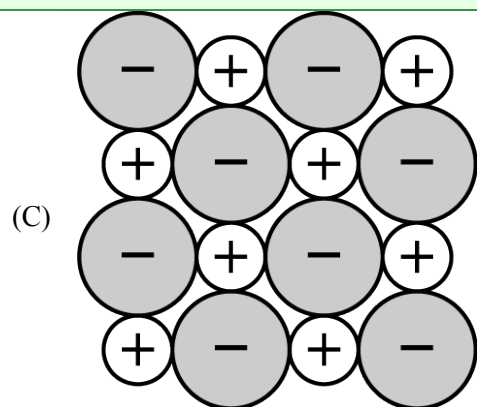
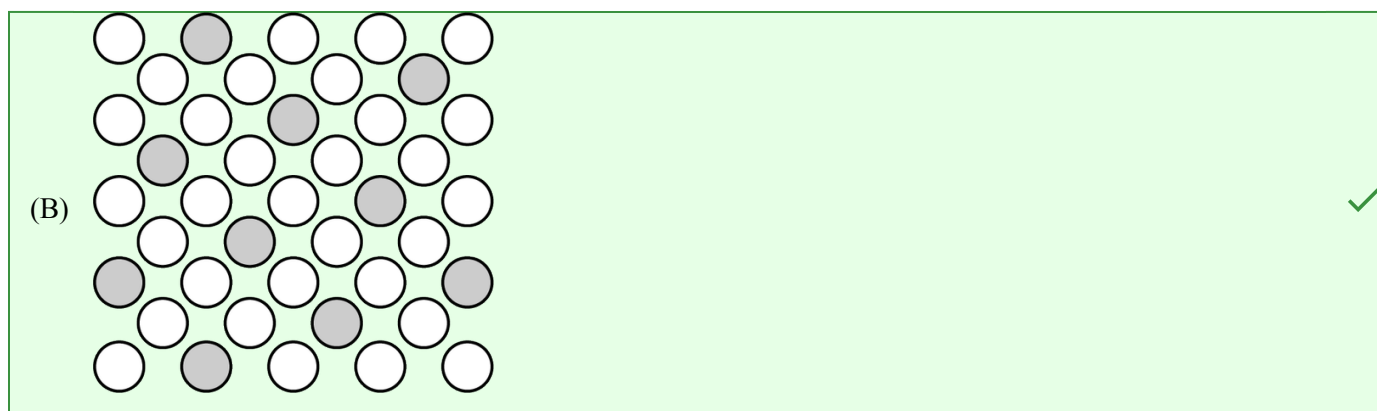
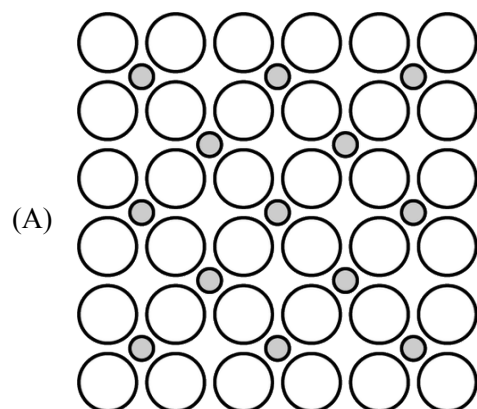


Chapter 03 Part 1: States and Properties

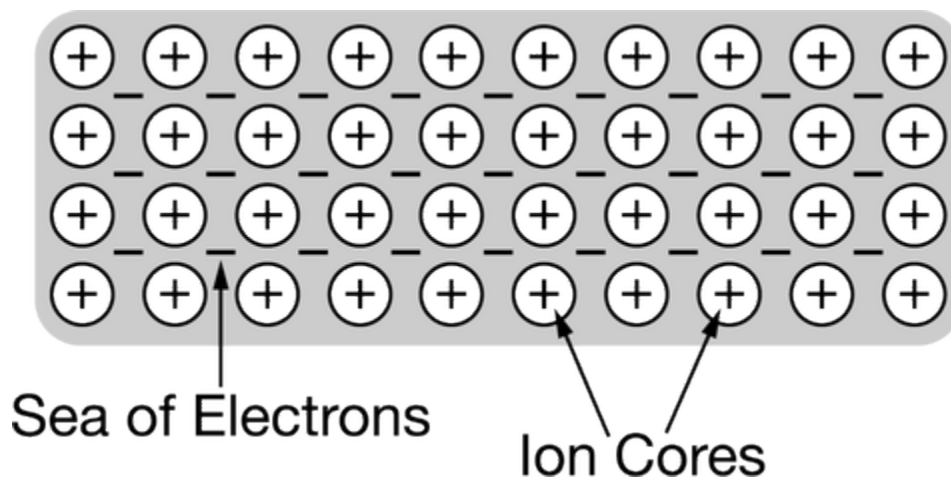
1. At 298 K and 1 atm, bromine is a liquid with a high vapor pressure, whereas chlorine is a gas. This provides evidence that, under these conditions, the
- (A) forces among Br₂ molecules are greater than those among Cl₂ molecules ✓
- (B) forces among Br₂ molecules are weaker than the Br—Br bond
- (C) forces among Cl₂ molecules are stronger than the Cl—Cl bond
- (D) Br—Br bond is stronger than the Cl—Cl bond
- (E) Br—Br bond is weaker than the Cl—Cl bond
2. Copper atoms and zinc atoms have the same atomic radius, 135 picometers. Based on this information, which of the following diagrams best represents an alloy containing only copper and zinc atoms?

Chapter 03 Part 1: States and Properties



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3.



A particle-level diagram of a metallic element is shown above. Typically, metals are both malleable and ductile. The best explanation for these properties is that the electrons involved in bonding among metal atoms are

- (A) unequally shared and form nondirectional bonds
- (B) unequally shared and form highly directional bonds
- (C) equally shared and form nondirectional bonds
- (D) equally shared and form highly directional bonds



4. A sample of a hard, solid binary compound at room temperature did not conduct electricity as a pure solid but became highly conductive when dissolved in water. Which of the following types of interactions is most likely found between the particles in the substance?

- (A) Ionic bonds
- (B) Metallic bonds
- (C) Covalent bonds
- (D) Hydrogen bonds



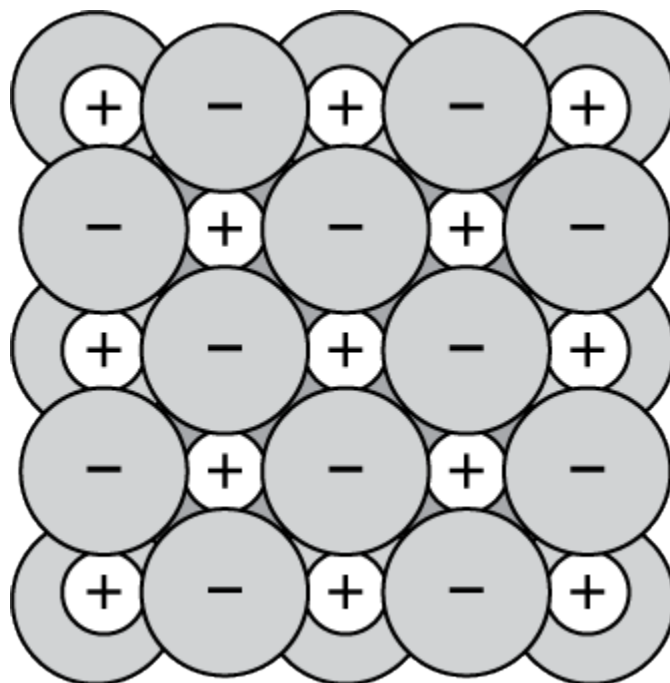
5. At room temperature $I_2(s)$ is a molecular solid. Which of the following provides a characteristic of $I_2(s)$ with a correct explanation?

- (A) It has a high melting point because it has weak intermolecular forces.
- (B) It is hard because it forms a three-dimensional covalent network.
- (C) It is not a good conductor of electricity because its valence electrons are localized in bonding and nonbonding pairs.
- (D) It is very soluble in water because its molecules are polar.



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6.

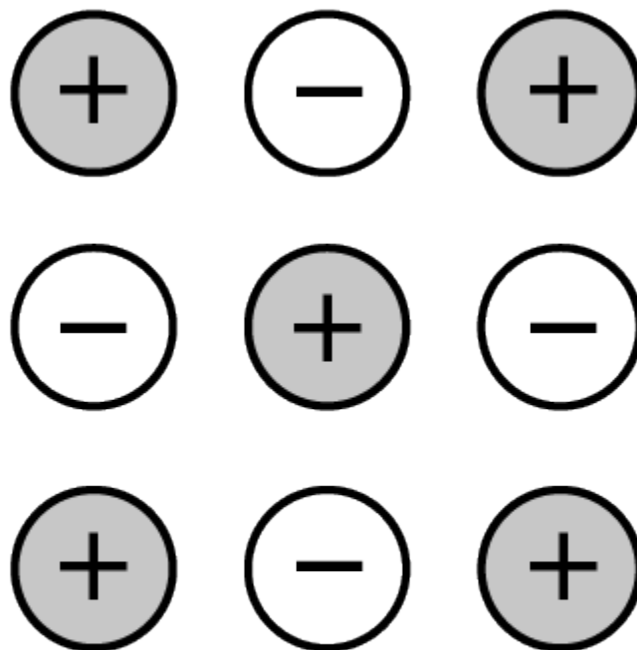


Which of the following correctly indicates whether the solid represented by the particulate model shown above conducts electricity and explains why or why not?

- (A) It conducts electricity because it is made of positive and negative ions.
- (B) It conducts electricity because it is made of particles of different sizes.
- (C) It does not conduct electricity because its ions cannot move freely within the solid. ✓
- (D) It does not conduct electricity because there are small spaces between the particles.

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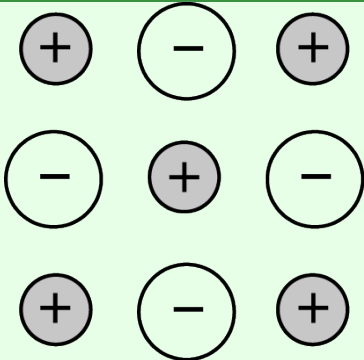
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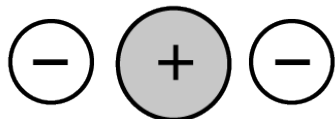
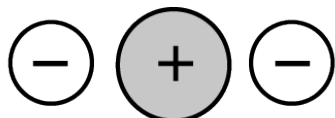


Solid KF

The particle-level diagram above represents the structure of solid KF. Although the molar mass of KCl is greater than that of KF, the density of KCl is actually less than that of KF. Which of the following representations of the structure of KCl best helps to explain this phenomenon?

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(A)  



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8.

Element	Metallic Radius (pm)	Melting Point (°C)	Common Oxidation State
Au	144	1064	1+, 3+
Cu	128	1085	1+, 2+
Ag	144	961	1+

To make Au stronger and harder, it is often alloyed with other metals, such as Cu and Ag. Consider two alloys, one of Au and Cu and one of Au and Ag, each with the same mole fraction of Au. If the Au/Cu alloy is harder than the Au/Ag alloy, then which of the following is the best explanation based on the information in the table above?

- (A) Cu has two common oxidation states, but Ag has only one.
- (B) Cu has a higher melting point than Au has, but Ag has a lower melting point than Au has.
- (C) Cu atoms are smaller than Ag atoms, thus they interfere more with the displacement of atoms in the alloy. ✓
- (D) Cu atoms are less polarizable than are Au or Ag atoms, thus Cu has weaker interparticle forces.

9. Of the following, the best explanation for the fact that most gases are easily compressed is that the molecules in a gas

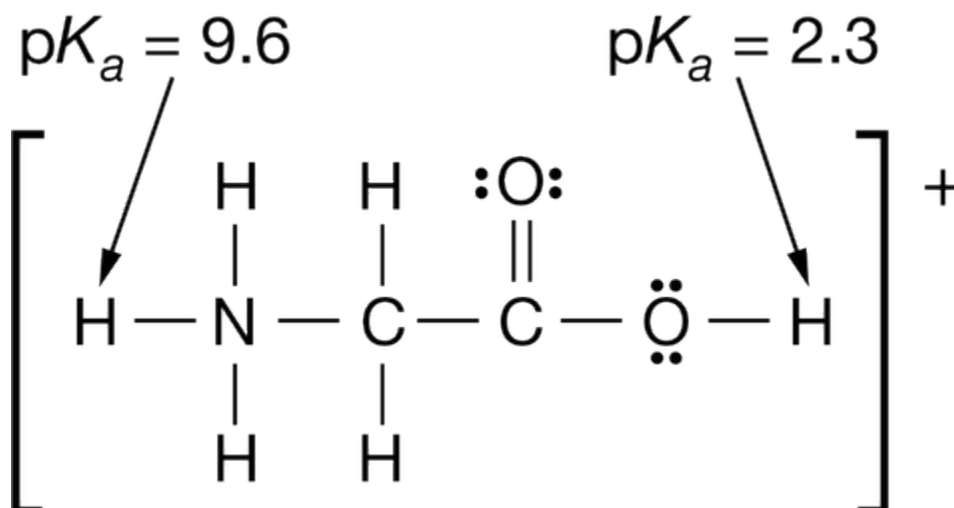
(A) are in constant motion

(B) are relatively far apart ✓

(C) have relatively small masses

(D) have a real, nonzero volume

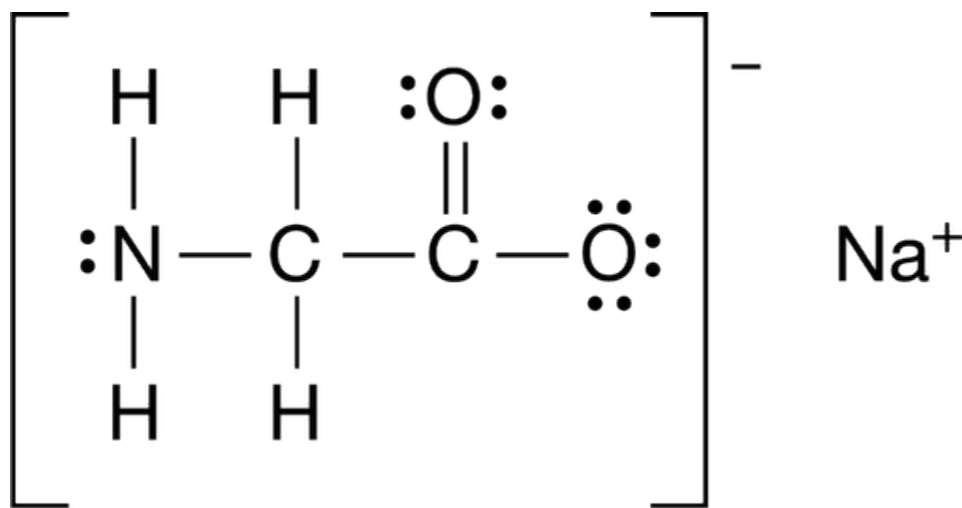
(E) move slower as temperature decreases



The structural formula of the glycinium cation is shown above. Arrows indicate the $\text{p}K_a$ values for the labile protons in the molecule.

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10.



Which of the following statements is true about sodium glycinate, represented above?

- (A) It is an ionic solid at room temperature. ✓
- (B) It is not soluble in water.
- (C) It evaporates easily because of the lack of hydrogen bonding.
- (D) It dissolves in water to form an acidic solution.

11. Which of the following could be the identity of a white crystalline solid that exhibits the following properties?

- It melts at 320°C.
- It does not conduct electricity as a solid.
- It conducts electricity in an aqueous solution.

(A) $\text{C}_6\text{H}_{12}\text{O}_6(s)$

(B) $\text{NaOH}(s)$ ✓

(C) $\text{SiO}_2(s)$

(D) $\text{Cu}(s)$

12.

Compound	Lattice Energy (kJ/mol)
NaF	930
MgF ₂	2978

The energy required to dissociate an ionic solid into gaseous ions (lattice energy) for the compounds NaF and MgF₂ is shown in the table above. On the basis of Coulomb's law, which of the following best helps to explain the large difference between the lattice energies of NaF and MgF₂ ?

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- (A) The solubility of MgF_2 is less than that of NaF .
- (B) The electronegativity of Mg is greater than that of Na .
- (C) The mass of the Mg cation is greater than that of the Na cation.

(D) The charge of the Mg cation is larger than that of the Na cation. ✓

13. A certain crystalline substance that has a low melting point does not conduct electricity in solution or when melted. This substance is likely to be

- (A) a covalent network solid
- (B) a metallic solid
- (C) a polymer
- (D) an ionic solid

(E) a molecular solid ✓

14. A solid is tested in a laboratory to determine its properties. Which of the following provides the best evidence to support the hypothesis that the solid is a metal as opposed to a nonmetal?

(A) It reacts with oxygen.

(B) It is lustrous and malleable. ✓

(C) It has a high melting point (1700°C).

(D) It has the same density as aluminum (2.7 g/mL).

15. A student is given a sample of a pure, white crystalline substance. Which of the following would be most useful in providing data to determine if the substance is an ionic compound?

(A) Examining the crystals of the substance under a microscope

(B) Determining the density of the substance

(C) Testing the electrical conductivity of the crystals

(D) Testing the electrical conductivity of an aqueous solution of the substance ✓

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16.

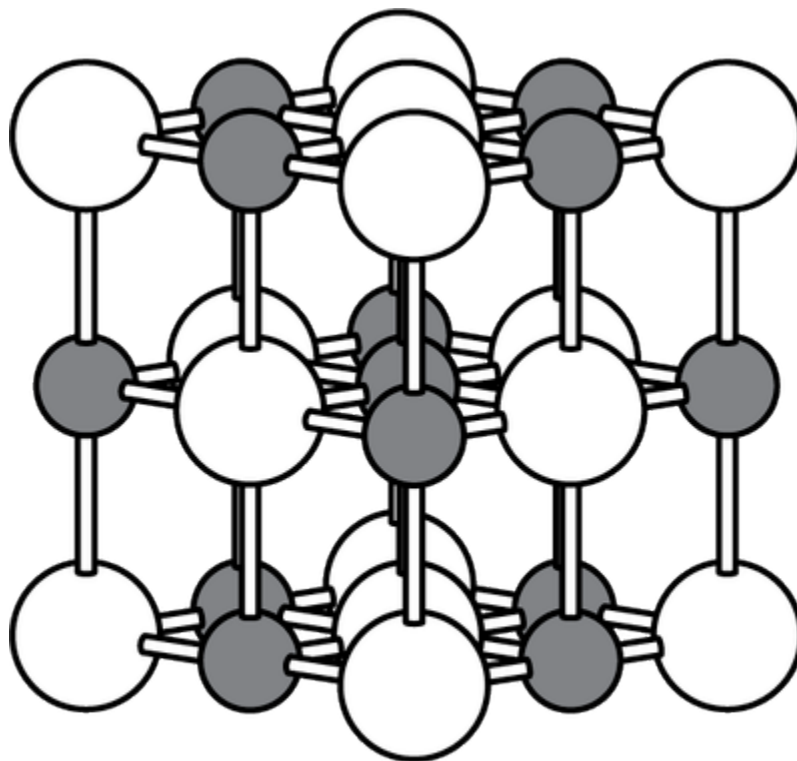


Diagram 1

Chapter 03 Part 1: States and Properties

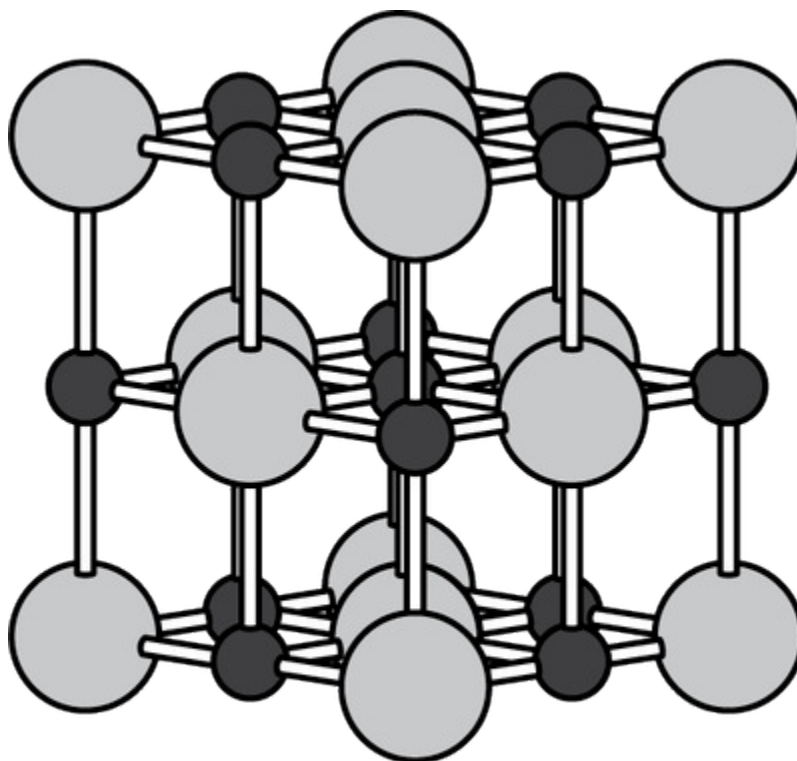


Diagram 2

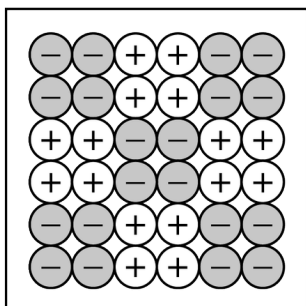
The ionic compounds NaCl and MgS are represented by the diagrams above. Which statement correctly identifies diagram 1 and identifies the compound with the lower melting point, explaining why?

- Diagram 1 represents NaCl; it has a lower melting point than MgS has because the coulombic attractions between the singly charged Na^+ ions and the Cl^- ions in NaCl are stronger than those between the ions in MgS.
- (A)
- (B) Diagram 1 represents NaCl; it has a lower melting point than MgS because the coulombic attractions between its singly charged Na^+ ions and the Cl^- ions are weaker than those between the ions in MgS. ✓
- Diagram 1 represents MgS; it has a lower melting point than NaCl because the coulombic attractions between its doubly charged Mg^{2+} ions and the S^{2-} ions are stronger than those between the ions in NaCl.
- (C)
- Diagram 1 represents MgS; it has a lower melting point than NaCl because the coulombic attractions between the doubly charged Mg^{2+} ions and the S^{2-} ions are weaker than those between the ions in NaCl.
- (D)

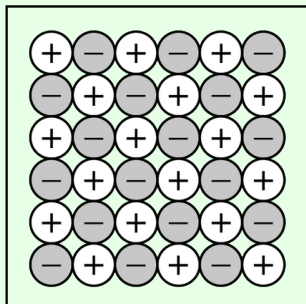
17. Of the following diagrams, which best represents the structure of solid KF ?

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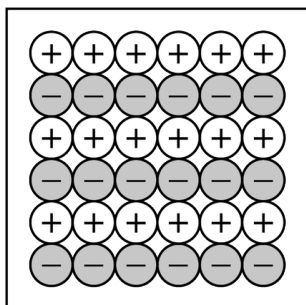
(A)



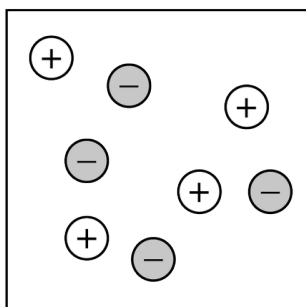
(B)



(C)

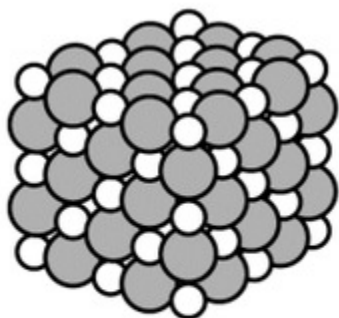


(D)

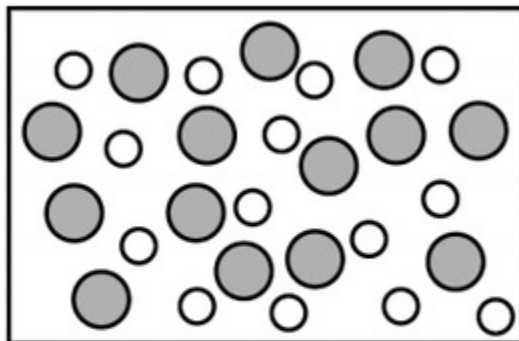


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18.



Solid MgO



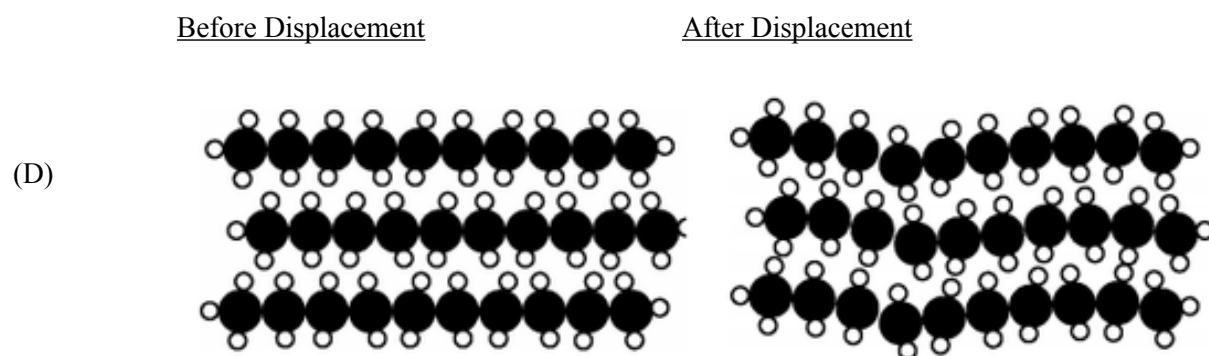
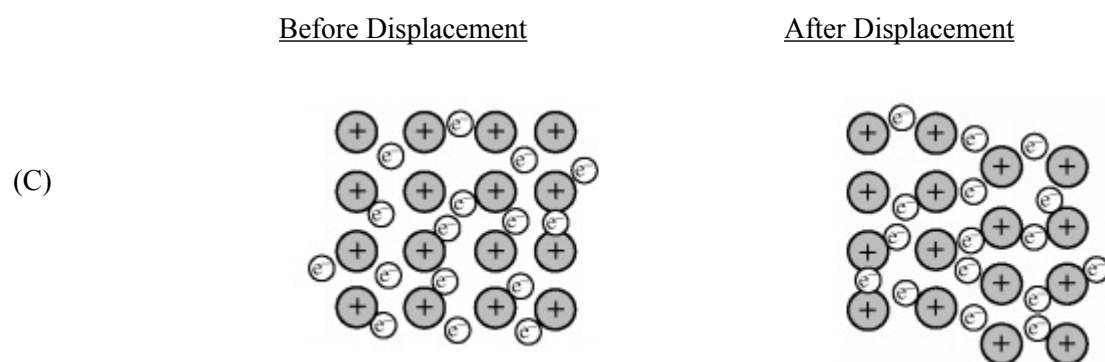
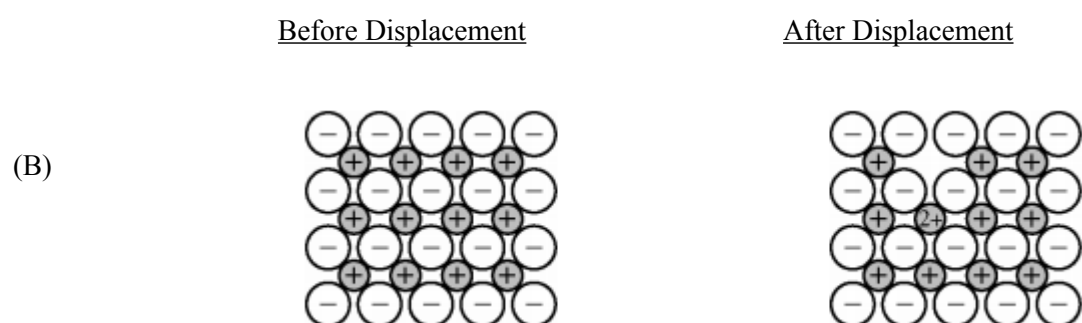
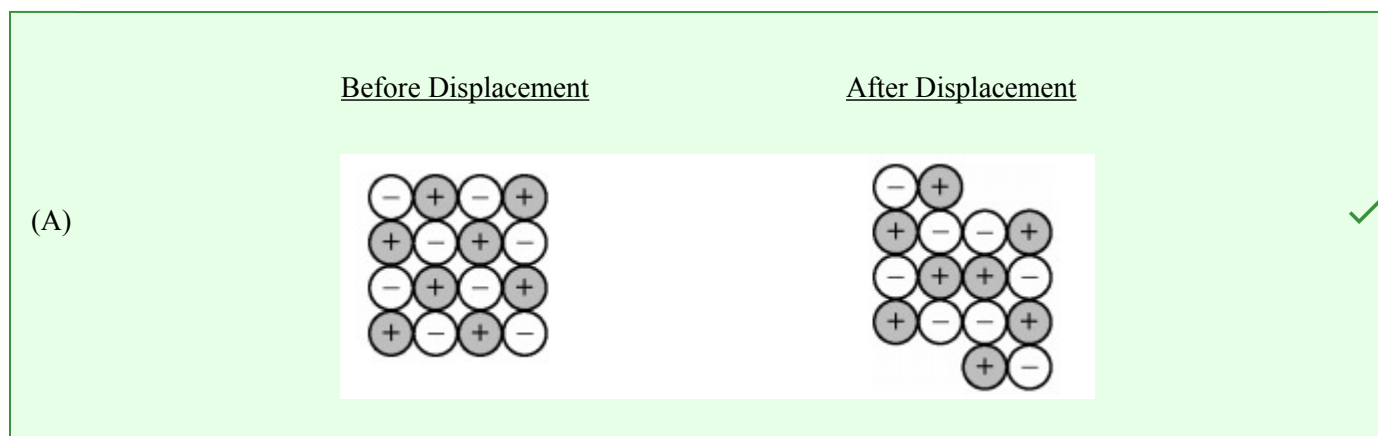
Liquid MgO

Based on the diagram above, which of the following best helps to explain why $\text{MgO}(s)$ is not able to conduct electricity, but $\text{MgO}(l)$ is a good conductor of electricity?

- (A) $\text{MgO}(s)$ does not contain free electrons, but $\text{MgO}(l)$ contains free electrons that can flow.
- (B) $\text{MgO}(s)$ contains no water, but $\text{MgO}(l)$ contains water that can conduct electricity.
- (C) $\text{MgO}(s)$ consists of separate Mg^{2+} ions and O^{2-} ions, but $\text{MgO}(l)$ contains MgO molecules that can conduct electricity.
- (D) $\text{MgO}(s)$ consists of separate Mg^{2+} ions and O^{2-} ions held in a fixed lattice, but in $\text{MgO}(l)$ the ions are free to move and conduct electricity. ✓

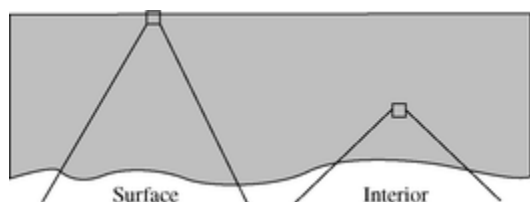
19. Which of the following diagrams best illustrates how a displacement in an ionic crystal results in cleavage and brittleness?

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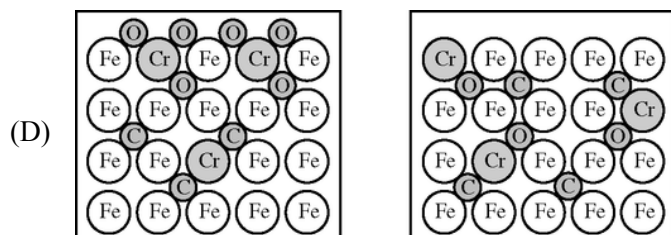
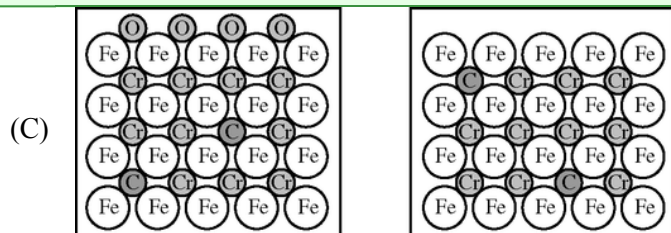
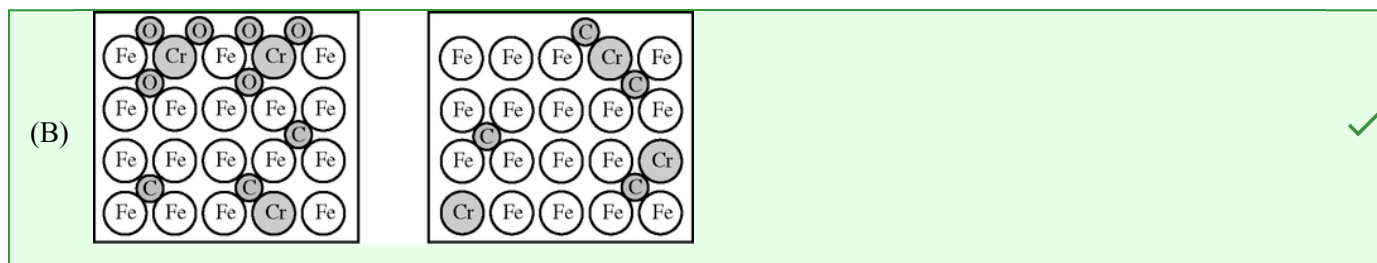
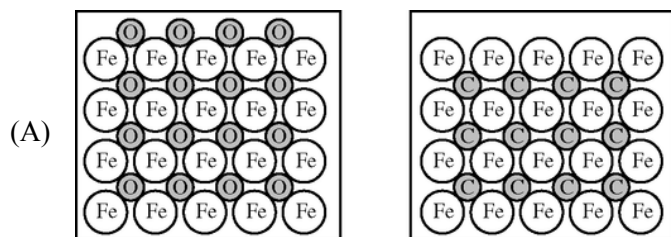


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20. Steel is an alloy consisting of Fe with a small amount of C. Elemental Cr can be added to steel to make the steel less likely to rust; Cr atoms react with oxygen in the air to form a nonreactive layer of chromium oxide on the surface of the steel, preventing the oxidation of underlying Fe atoms. A sample of steel-chromium alloy contains 15 percent Cr by mass. Which of the following diagrams best shows a particle-level view of a surface section and an interior section of the alloy represented below at the left? (The atomic radii of the atoms involved are given in the table below at the right.)

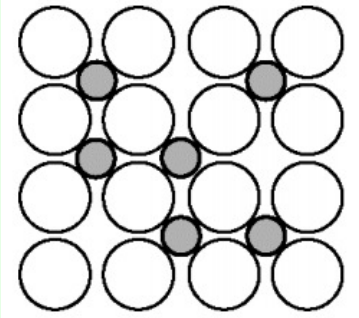


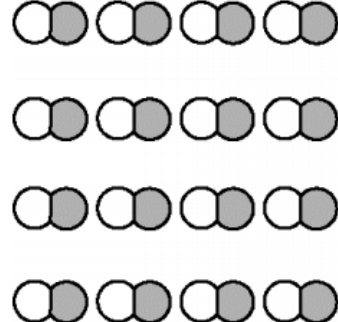
Element	Molar Mass (g/mol)	Atomic Radius (pm)
Fe	55.85	125
Cr	52.00	127
C	12.01	77
O	16.00	73

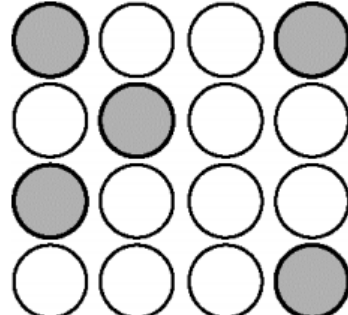


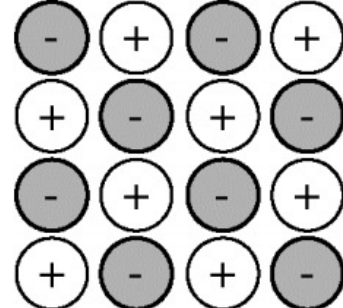
21. Which of the following diagrams best depicts an alloy of Ni and B?

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(A) 

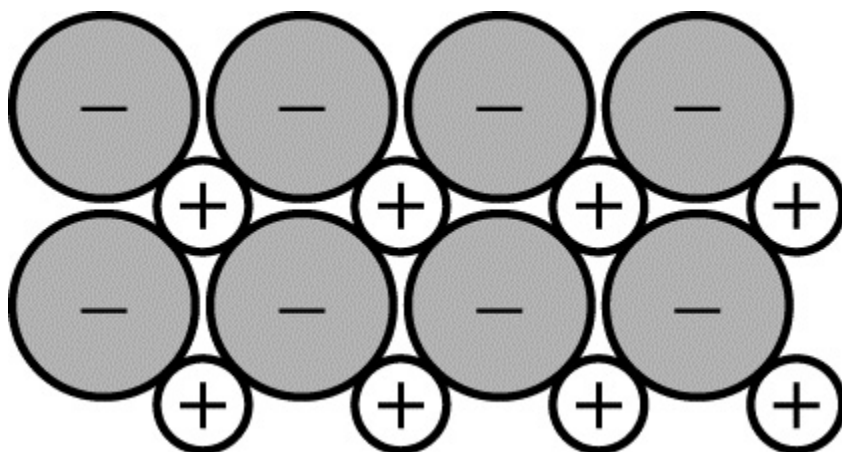
(B) 

(C) 

(D) 

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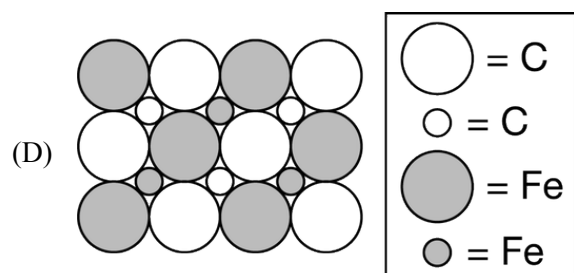
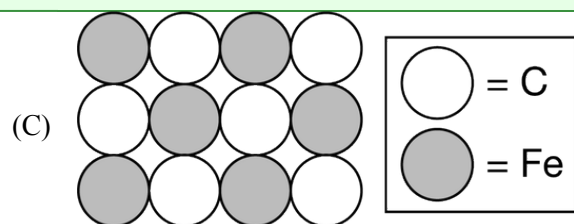
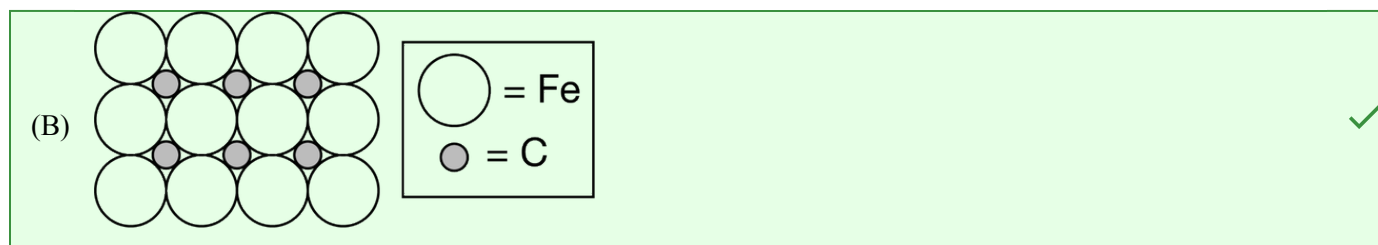
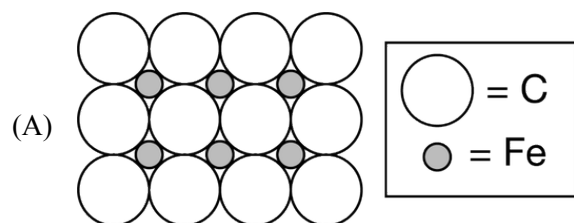
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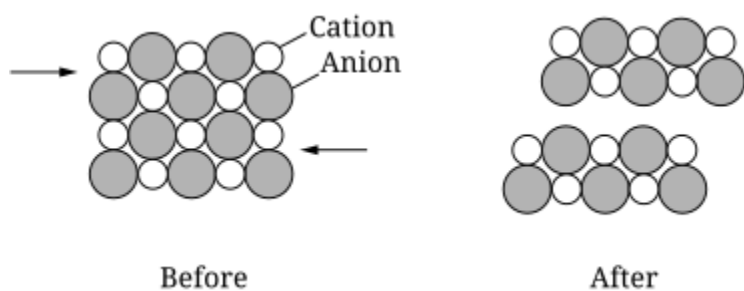
Which of the following most likely describes the solid represented in the diagram above?

- (A) It is soft and is a poor thermal and electrical conductor.
 - (B) It has a low melting point and is a poor thermal and electrical conductor as a solid.
 - (C) It is a brittle, water-soluble electrolyte that is a poor thermal and electrical conductor as a solid. ✓
 - (D) It is malleable and is an excellent thermal and electrical conductor as a solid.
23. Steel is an alloy containing Fe atoms and C atoms. Which of the following diagrams best represents the particle-level structure of steel?

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24.



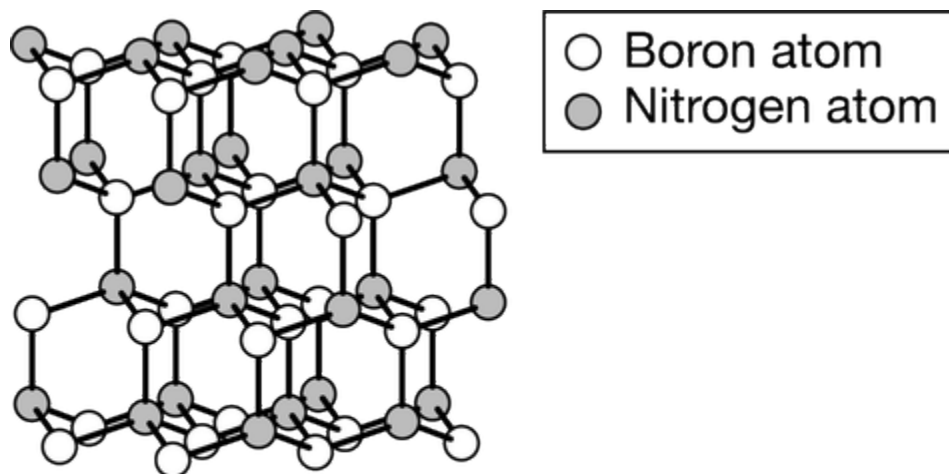
When a stress is applied to an ionic crystal, the planes of the ions slide and separate due to the alignment of like charges, as shown in the diagram above. Which of the following characteristics of the crystal is represented by the observation?

- (A) Luster
- (B) Brittleness
- (C) High melting point
- (D) Solubility in water

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25. The best explanation for the fact that diamond is extremely hard is that diamond crystals
- (A) are made up of atoms that are intrinsically hard because of their electronic structures
 - (B) consist of positive and negative ions that are strongly attracted to each other
 - (C) are giant molecules in which each atom forms strong covalent bonds with all of its neighboring atoms ✓
 - (D) are formed under extreme conditions of temperature and pressure
 - (E) contain orbitals or bands of delocalized electrons that belong not to single atoms but to each crystal as a whole

26.

Cubic Form of
Boron Nitride

The structure of one form of boron nitride is represented above. This form of boron nitride is one of the hardest substances known. Which of the following best helps explain why boron nitride is so hard?

- (A) Boron ions and nitrogen ions are held together by ionic bonds.
- (B) Boron nitride is a network solid of atoms connected by covalent bonds with fixed bond angles. ✓
- (C) Boron nitride is an alloy, and alloys are typically harder than the elements used to make them.
- (D) Boron nitride is a polymer made of long chains of boron atoms and nitrogen atoms held together by dispersion forces.

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27.

Substance	X	Y	Z
Melting Point ($^{\circ}\text{C}$)	1600	650	1418
Electrical Conductivity in Solid Phase	None	High	None
Electrical Conductivity in Liquid Phase	None	High	High

Three substances were studied in the laboratory, and the data in the table above were collected. Based on the data, which of the following shows the type of bonding in each substance?

(A)

<u>Substance X</u>	<u>Substance Y</u>	<u>Substance Z</u>
Network covalent	Ionic	Metallic

(B)

<u>Substance X</u>	<u>Substance Y</u>	<u>Substance Z</u>
Ionic	Molecular	Network covalent

(C)

<u>Substance X</u>	<u>Substance Y</u>	<u>Substance Z</u>
Molecular	Network covalent	Metallic

(D)

<u>Substance X</u>	<u>Substance Y</u>	<u>Substance Z</u>
Network covalent	Metallic	Ionic



28. In which of the following processes are covalent bonds broken?

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- (A) $\text{I}_2(s) \rightarrow \text{I}_2(g)$
 (B) $\text{CO}_2(s) \rightarrow \text{CO}_2(g)$
 (C) $\text{NaCl}(s) \rightarrow \text{NaCl}(l)$

(D) $\text{C}(\text{diamond}) \rightarrow \text{C}(g)$ ✓

(E) $\text{Fe}(s) \rightarrow \text{Fe}(l)$

29.

Type of Steel	% Carbon	Characteristics	Uses
Low-carbon steel	< 0.2 %	Malleable and ductile	Chains and nails
High-carbon steel	0.6 – 1.5 %	Hard and brittle	Cutting tools

The table above provides some information about two types of steel, both of which are alloys of iron and carbon. Which of the following best helps to explain why high-carbon steel is more rigid than low-carbon steel?

- (A) Elemental carbon is harder than elemental iron.
 (B) The additional carbon atoms within the alloy make the high-carbon steel less dense.
 (C) The additional carbon atoms within the alloy increase the thermal conductivity of the high-carbon steel.

(D) The additional carbon atoms within the alloy make it more difficult for the iron atoms to slide past one another. ✓

30.

Compound	Melting Point (° C)
MgO	2852
NaF	993

The table above shows the melting points of $\text{MgO}(s)$ and $\text{NaF}(s)$. Which of the following best helps explain why the melting point of $\text{MgO}(s)$ is much higher than that of $\text{NaF}(s)$?

- (A) The mass of F^- ions is greater than that of O^{2-} ions.
 (B) The mass of Mg^{2+} ions is greater than that of Na^+ ions.
 (C) The difference between the electronegativity values of Mg and O is smaller than the difference between the values of Na and F.

(D) The charges of Mg^{2+} and O^{2-} ions are greater than those of Na^+ and F^- ions. ✓

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31.

	NaF	MgO
Boiling Point (°C)	1695	3600

	Na ⁺	Mg ²⁺	F ⁻	Cl ⁻	O ²⁻
Ionic Radius (pm)	76	72	133	181	140

Based on the data in the tables above, which of the following statements provides the best prediction for the boiling point of NaCl ?

- (A) NaCl will have a lower boiling point than NaF because the coulombic attractions are weaker in NaCl than in NaF . ✓
- (B) NaCl will have a boiling point between that of NaF and MgO because the covalent character of the bonds in NaCl is intermediate between that of MgO and NaF.
- (C) NaCl will have a higher boiling point than MgO because the ions are spaced farther apart in NaCl .
- (D) NaCl will have a higher boiling point than MgO because the energy required to transfer electrons from the anion to the cation is larger in NaCl than in MgO .

32. At 298 K and 1 atm, Br₂ is a liquid with a high vapor pressure, and Cl₂ is a gas. Those observations provide evidence that under the given conditions, the

- (A) forces among Br₂ molecules are stronger than those among Cl₂ molecules ✓
- (B) forces among Cl₂ molecules are stronger than the Cl—Cl bond
- (C) Br—Br bond is stronger than the Cl—Cl bond
- (D) Cl—Cl bond is stronger than the Br—Br bond

33. Silicon crystals are semiconductors. Which of the following is a correct reason for the increase in the conductivity of Si crystals when a small fraction of Si atoms are replaced with those of a different element?

- (A) P atoms introduce additional mobile negative charges. ✓
- (B) P atoms introduce additional mobile positive charges.
- (C) Ge atoms have more electrons than Si atoms have.
- (D) Ge atoms are much smaller than Si atoms.